

A2Q+: Improving Accumulator-Aware Weight Quantization

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Abstract

Quantization techniques commonly reduce the inference costs of neural networks by restricting the precision of weights and activations. Recent studies show that also reducing the precision of the accumulator can further improve hardware efficiency at the risk of numerical overflow, which introduces arithmetic errors that can degrade model accuracy. To avoid numerical overflow while maintaining accuracy, recent work proposed accumulator-aware quantization (A2Q)—a quantization-aware training method that constrains model weights during training to safely use a target accumulator bit width during inference. Although this shows promise, we demonstrate that A2Q relies on an overly restrictive constraint and a sub-optimal weight initialization strategy that each introduce superfluous quantization error. To address these shortcomings, we introduce: (1) an improved bound that alleviates accumulator constraints without compromising overflow avoidance; and (2) a new strategy for initializing quantized weights from pre-trained floating-point checkpoints. We combine these contributions with weight normalization to introduce A2Q+. We identify and characterize the various trade-offs that arise as a consequence of accumulator constraints and support our analysis with experiments that show A2Q+ significantly improves these trade-offs when compared to prior methods.

1. Introduction

Quantizing neural network weights and activations to low-precision integers can drastically reduce the inference costs of multiplications. However, the resulting products are commonly accumulated at high-precision and thus require high-precision additions and registers. Recent studies show that reducing the standard 32-bit accumulators to 16 bits can

yield a near-optimal $2\times$ increase in throughput and bandwidth efficiency on ARM processors (de Bruin et al., 2020; Xie et al., 2021) and up to a $1.6\times$ reduction in resource utilization on FPGAs (Colbert et al., 2023). However, exploiting such an optimization is highly non-trivial in practice as doing so also incurs a high risk of numerical overflow, which introduces arithmetic errors that can significantly degrade model accuracy (Ni et al., 2021).

To train quantized neural networks (QNNs) for low-precision accumulation, Colbert et al. 2023 recently proposed accumulator-aware quantization (A2Q). While prior approaches had sought to either reduce the risk of numerical overflow (Xie et al., 2021; Li et al., 2022; Azamat et al., 2022) or mitigate its impact on model accuracy (Ni et al., 2021; Blumenfeld et al., 2023), A2Q circumvents arithmetic errors caused by numerical overflow by constraining model weights to restrict the range of outputs. In doing so, A2Q provides state-of-the-art performance for low-precision accumulation with guaranteed overflow avoidance.

Our work contributes to this body of research by further improving the trade-off between accumulator bit width and model accuracy. We show that A2Q relies on: (1) an overly restrictive ℓ_1 -norm bound that constrains QNNs more than necessary; and (2) a sub-optimal initialization strategy that forces QNNs to recover from superfluous quantization error. In addressing these shortcomings, we establish a new state-of-the-art for low-precision accumulation with guaranteed overflow avoidance. Our results show for the first time that ResNet50 (He et al., 2016) can maintain 95% of its baseline accuracy when trained on ImageNet (Deng et al., 2009) to accumulate at 12 bits without overflow, resulting in a +17% improvement in test top-1 accuracy over A2Q.

Our contributions are four-fold: (1) we introduce a new theoretical analysis for an improved ℓ_1 -norm bound that alleviates accumulator constraints without compromising overflow avoidance; (2) we introduce a weight initialization strategy that minimizes the initial weight quantization error caused by accumulator constraints; (3) we combine (1) and (2) with weight normalization (Salimans & Kingma, 2016) to introduce A2Q+ and show significant improvements in the trade-off between accumulator bit width and model accuracy; and (4) we identify and characterize various trade-offs that arise as a consequence of accumulator constraints.

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